No of Pages :2 **Course Code : 23MXAM**

Roll No:
(To be filled in by the candidate)

PSG COLLEGE OF TECHNOLOGY, COIMBATORE - 641 004

SEMESTER EXAMINATIONS, NOVEMBER 2024

MCA Semester: 2

23MXAM OPEN SOURCE SYSTEMS

Time: 3 Hours **Maximum Marks :100**

INSTRUCTIONS:
1. Answer **ALL** questions. Each question carries 25 Marks.

1. a) State the implications of open-source systems on software development.

   b) i) Why does open-source software also require software licensing? State the significance of such licenses.

   ii) Explain the steps involved in software testing and bug fixing before release of any software.

   c) i) How do the Bazaar model of software development help the open-source software evolve and help the development community than the cathedral model of software development.

   (OR)

   ii) Describe the various types of licenses available under software licensing. Compare all the licenses available for Software and open-source software. How these licensing does differ with copyrights? Give proper examples.

2. a) Why application developers prefer to use R software? Justify with proper reasons.

   b) i) Write R code to read a matrix and determine its row sum, column sum and sum of all elements.

   ii) Differentiate between apply() and lapply() functions in R. Determine the output generated by the following R coding.

   c) matrix_data<- matrix(1:9, nrow = 3, byrow = TRUE)
   apply(matrix_data, 2, sum) # 2 indicates columns

   Describe in detail how the data files could be read from csv file in R language and also write the code for reading and updating a csv file using R language.

3. a) Compare and contrast NoSQL and SQL on three main features.

   b) i) How MongoDB keep availability and consistency over network error and detachment happen.

   ii) Describe the proper methods and MongoDB code to do insert, delete and update the DB using JSON objects. Give proper examples.

c) With an example of your choice, explain how two-phase commit protocol could maintain consistency and transactional integrity in distributed NoSQL databases.

4. a) How does Selenium kind of Open-source software help the development team?

b) i) State and explain the three main files that Bootstrap framework provides for developers.

ii) State and explain how Joomla offers web content management.

c) Describe the architecture and working of Git as a version control system. Also explain the different storage areas managed by the Git.

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FD/VS